

Multi-Scale Adaptive Inspection Framework for Voids in MOSFETs

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Abstract—During reflow soldering, Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) are prone to internal voids. A high void ratio for a MOSFET significantly affects its electrical performance to further decrease the reliability and lifespan of the electronic device. In this article, a multi-scale adaptive inspection framework is proposed to inspect internal voids in MOSFETs through X-ray imaging. First, a location-aware vertex detection scheme is utilized to detect the transistor region possibly involving the voids, which makes full advantages of the neighborhood grayscale distributions of the transistor region vertices. Then, after the detected transistor region is preprocessed and divided into several patches, coarse-to-fine void inspection is performed on the patches for void inspection. Specifically, a multi-scale scheme is proposed for coarse void inspection to identify whether the patches involve the potential voids, in which an identification function is defined by integrating the constructed multi-scale radial decaying weight matrix and a fitting metric to characterize the patches. An adaptive thresholding scheme is proposed to finely inspect the voids in the MOSFETs, in which the segmentation threshold model is constructed based on grayscale distributions of the MOSFET X-ray images. Finally, the void ratio is calculated according to the detected transistor and the voids. Experimental results indicate that the proposed framework achieves better inspection performance for voids in MOSFETs than some existing methods at the speed of 0.812s per MOSFET, with an average Dice of 0.8852, an Accuracy of 97.11%, and an F1-score of 0.9819.

Index Terms—MOSFET, void inspection, location-aware vertex detection, multi-scale scheme, grayscale distribution.

I. INTRODUCTION

MOSFET, as a kind of power semiconductor device, is widely used in electronic fields such as high-frequency switching power supplies, power amplifiers, and motor drive

systems, due to its advantages of high voltage tolerance, low conduction voltage, high switching speed, and high efficiency [1], [2]. During the packaging process, MOSFETs are prone to internal voids due to trapped gases, flux residues, and reflow soldering issues [3]. These voids increase the thermal resistance and significantly reduce the heat dissipation performance, which affect the reliability and lifespan of the device [4], [5], [6]. Therefore, it is essential to inspect internal voids in MOSFETs during industrial production, which is commonly implemented by manual inspection through X-ray technology [7], [8]. However, a long-term manual inspection not only increases the risk of eye strain and related diseases due to visual fatigue, but also leads to unacceptable errors in judgment [9], [10].

Deep learning has been applied in the industrial inspection field, due to its powerful feature extraction and learning capabilities [11]. Due to its data-driven nature, a large number of defect samples are required to train a promising model for inspection [12], [13]. However, in practical industrial production, defect samples are not only difficult to collect [14], but the inspection model needs to be retrained when new samples appear [15]. In addition, the labor-intensive and costly manual annotation of defect samples limits the application of deep learning in industrial scenarios [16]. Thus, despite the advances in deep learning, traditional image processing methods still retain an industrial significance for inspection, especially when defect samples are scarce or the retraining of the model is difficult.

In recent years, traditional image processing has been widely applied to automated inspection of voids in electronic components to deal with the issues of manual inspection. Said et al. [17] combined template matching and Laplacian of Gaussian (LoG) to inspect the voids in the Ball Grid Array (BGA) solder joints, in which template matching was utilized to detect solder joint regions based on the shape, area, and distribution characteristics and the LoG operator was used for void inspection. Peng and Nam [18] proposed Blob filtering with mean box filters of different sizes to hierarchically inspect voids of various sizes in BGA solder joints. Le et al. [19] combined several traditional image processing methods, such as grayscale morphology, histogram stretching, fuzzy enhancement, and Blob analysis, with interactive ray profile extraction to inspect voids in BGA solder joints. Mouri et al. [20] inspected voids through non-negative matrix factorization based on the assumption that voids in BGA solder joints were sparse variables. Nuanprasert et al. [21] used

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image mapping to localize the Regions of Interest (ROIs) in BGA solder joints and enhanced the contrast through adaptive histogram equalization. Subsequently, the LoG operator was applied to inspect the enhanced ROI for void localization. Ahuja and Neeluru [22] used adaptive thresholding to detect the ROIs of BGA solder joints, and then combined the LoG operator with radial and angular scanning to inspect voids within the ROI. Xiao et al. [23] obtained the binary image of BGA using maximum variance thresholding, then reconstructed the image through morphological processing, and finally inspected voids using image subtraction. Eom and Min [24] segmented each BGA solder ball region into one or more subregions based on grayscale values, and then inspected voids using local variance-weighted local thresholding. Xiao et al. [25] divided the chip resistor image into two regions and performed coarse inspection of voids in both regions using a custom multi-scale circular convolution kernel and a circular-prior-based local pre-fitting (LPF) active contour model. Subsequently, based on an average grayscale strategy and a custom shape factor, they performed fine inspection of voids.

Although above previous methods have performed fair performance on void inspection for some electronic components, such as BGA and chip resistor, they can possibly meet with some difficulties to inspect internal voids in MOSFETs, since the voids in different electronic components demonstrate different image characteristics. Since some solders overflow outside the transistor regions of the soldered MOSFETs, the edges of transistor regions are hardly localized in the acquired X-ray images, which means that previous thresholding methods utilized for BGA void inspection cannot well localize the edges of transistor regions, i.e., ROIs. Also, the voids in the MOSFETs demonstrate various sizes and various irregular shapes, which indicates that circular convolution kernels and circular priorities in [25] are not suitable for MOSFET void inspection.

To this end, a multi-scale adaptive inspection framework is designed to well inspect the irregular voids with various sizes in MOSFETs, involving ROI detection, void detection, and void ratio calculation. To well detect the transistor region (i.e., ROI), a location-aware vertex detection is utilized to localize its four vertices. Then, the ROI is divided into several patches that are preprocessed by several traditional image processing methods. Next, a coarse void inspection scheme is designed to identify the patches containing voids, abbreviated as MRDWM-FM, which seamlessly integrates the designed multi-scale radial decaying weight matrix and a fitting metric. Subsequently, an adaptive thresholding scheme, abbreviated as GDE-AT, is proposed based on grayscale distribution estimation to finely inspect the voids. Finally, the void ratio is calculated according to the detected ROI and the finely inspected voids in the MOSFET.

In summary, our contributions can be summarized as follows.

- 1) A multi-scale adaptive inspection framework is designed to automatically inspect the voids in MOSFETs, which includes location-aware vertex detection for transistor region localization, MRDWM-FM for coarse void

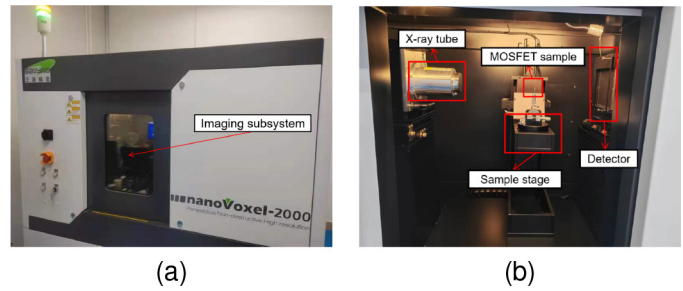


Fig. 1. Data acquisition system. (a) Industrial CT. (b) Imaging subsystem.

inspection, GDE-AT for fine void inspection, and void ratio calculation.

- 2) MRDWM-FM is proposed to perform coarse void inspection, which identifies the patches containing voids based on the scale and grayscale distribution characteristics of the patches. Specifically, a multi-scale radial decaying weight matrix and a fitting metric based on KL divergence are designed to extract the scale and grayscale distribution characteristics, respectively.
- 3) GDE-AT is proposed to finely inspect the voids in the patches containing voids, which constructs an adaptive thresholding function based on the grayscale distribution of each corresponding patch. Specifically, the constructed function is composed of a mean term, a standard deviation term, and a correction term, which are estimated based on the cumulative frequency of the grayscale values.

II. DATA ACQUISITION SYSTEM AND DATASET

A. Data Acquisition System

Due to the non-destructive nature of X-ray imaging technology for internal inspection in the electronic fields [26], an industrial CT (Tianjin Sanying nanoVoxel-2000) equipment shown in Fig. 1 was used to acquire X-ray images of MOSFETs. The imaging subsystem [Fig. 1(b)] is composed of an X-ray tube, a sample stage, a detector, and a computer numerical control (CNC) subsystem. The X-ray tube operates at a voltage of 110kV and a power of 16W. The detector has a resolution of 1536×1536 pixels, with a detector element size of $85\mu\text{m}$. The CNC subsystem features the following motion ranges: the detector has a travel range of 300mm along both the X-axis and Y-axis, while the sample stage has travel ranges of 500mm and 250mm along the X-axis and Z-axis, respectively. The X-ray tube emits X-ray that pass through the sample, with varying degrees of attenuation. Then, the attenuated X-ray eventually reaches the detector. The detector captures the differences in X-ray intensity and converts them into digital signals. The digital signals are next processed by the software NanoVoxelScan version 2.3 to generate X-Ray images that reflect the internal structure of the sample.

B. Dataset

In our study, 519 MOSFETs were utilized to acquire corresponding X-ray images by means of the industrial CT in Fig. 1, each of which has the size of 1536×1536 pixels.

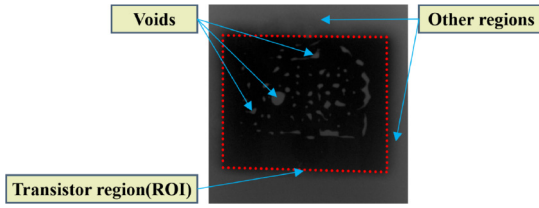


Fig. 2. X-ray MOSFET image.

To obtain the ground-truth void labels for MOSFET X-ray images, two quality control (QC) workers with over 5-year experience were invited to perform manual annotation. Then, an industrial expert with over 15 years of domain experience reviewed and verified the annotations. The Cohen's Kappa coefficient for annotations by two QC workers exceeds 0.95 at minimum and averages over 0.98, indicating a high inter-annotator agreement. Due to the commercial confidentiality and business privacy, the dataset cannot be publicly released. Fig. 2 shows an example of the X-ray images of a MOSFET. Some voids with various sizes and various irregular shapes emerge in the transistor region (outlined by a red dashed line), i.e., ROI. Due to the transistor region being primarily composed of metallic and semiconductor materials, it attenuates X-rays more than other regions, resulting in lower grayscale values. Some soldering lead overflowed into the transistor region, resulting that the edges of the transistor region are not well-distinct, which makes the localization of the ROI challenging. If the voids emerge in the ROI, the voids attenuate less X-rays than the non-voids in the ROI, which results that the voids demonstrate lighter than the non-voids.

III. METHODOLOGY

A. Proposed Framework

Fig. 3 illustrates our proposed multi-scale adaptive framework for MOSFET void inspection, which involves ROI detection, void detection, and void ratio calculation. In the ROI detection stage, as the ROI demonstrates approximately quadrilateral, a location-aware vertex detection scheme is proposed to detect the ROI in the acquired X-ray MOSFET image, which is the transistor region. In the void detection stage, the detected ROI goes through preprocessing via several operations to be divided into multiple patches. Next, the proposed MRDWM-FM performs on the enhanced patches to coarsely identify whether the patches involve voids, which combines the constructed multi-scale radial decaying weight matrix and a fitting metric for patch identification. Finally, the proposed GDE-AT is utilized to finely inspect the voids in the coarsely-identified patches containing voids, which combines a prior threshold model and grayscale distribution estimation for fine void inspection. In the void ratio calculation stage, the void ratio is calculated based on the pixel counts of the detected ROI and those of the detected voids in each X-ray MOSFET image, formulated as

$$V_{ratio} = \frac{S_V}{S_R} \times 100\% \quad (1)$$

where S_V and S_R represent the areas (i.e., pixel counts) of the voids and the ROI in the X-ray MOSFET image, respectively.

B. Vertex Detection

As shown in Fig. 4, the detected ROI has four vertices, which motivates us to detect it through vertex detection. However, as illustrated in Fig. 4, the vertices demonstrate dim in the image. Thus, a location-aware vertex detection scheme is proposed to transform vertex detection into an iterative optimization problem, described below.

For simplicity, the proposed vertex detection scheme is outlined by using the bottom-right vertex of the ROI. The coordinates of the bottom-right vertex (x_{br}, y_{br}) can be calculated by solving the following optimization problem

$$\begin{aligned} (x_{br}, y_{br}) &= \arg \min J(x_c, y_c) \\ \text{s.t.} \quad &\frac{3N_I}{4} < x_c, y_c \leq N_I - \frac{N_w-1}{2} \end{aligned} \quad (2)$$

$$J(x_c, y_c) = \frac{\sigma_{w_1}^2 + \sigma_{w_2}^2}{\sigma_w^2} + \alpha \times \frac{\mu_{w_1}}{\mu_{w_2}} \quad (3)$$

where N_w represents the size of the window, and N_I denotes the size of the X-ray MOSFET image. w and (x_c, y_c) represent the sliding window (blue box in Fig. 4) and its center, respectively; w_1 and w_2 represent the upper-left quarter of w (red dashed box) and the other region of w , respectively. σ_{w_1} , σ_{w_2} and σ_w^2 represent the variances of grayscale values in w_1 , w_2 and w , respectively; μ_{w_1} and μ_{w_2} represent the grayscale means of w_1 and w_2 , respectively. α is the balancing factor, empirically set to 1. The coordinates of the bottom-right vertex can be obtained by solving (2) using a uniform window traversal method. The coordinates of the other three vertices can also be detected in the similar manner. When the four vertices are detected, the ROI is detected according to the vertices.

C. Data Preprocessing

Since the ROI involves uneven grayscale distribution to possibly influence void inspection, it is better to partition it into several patches to alleviate this unwelcome influence. To do so, the ROI is successively preprocessed by 3×3 mean filtering to reduce the impact of noise, and Contrast Limited Adaptive Histogram Equalization (CLAHE) with a Clip Limit of 4 and a Tile Grid size of (28, 28) to enhance the image contrast. Then, the contrast-enhanced ROI is partitioned by a multi-segment partitioning method. Specifically, the coordinates of the n equidistant points are calculated for each of the four edges of the ROI. Subsequently, $(n-1)$ line segments are drawn connecting the corresponding equidistant points on the opposite edges, and each of these line segments is further divided into n equal parts to compute their respective equidistant points. This process results in a total of $(n+1) \times (n+1)$ grid points. Finally, each group of four adjacent grid points forms a patch, indicating that $n \times n$ patches are achieved for the ROI. To address the uneven gray scale distribution within the ROI while avoiding patches that consist entirely of voids, n is empirically set to 3. To mitigate boundary issues, each patch is empirically expanded along each edge with the 2-pixel

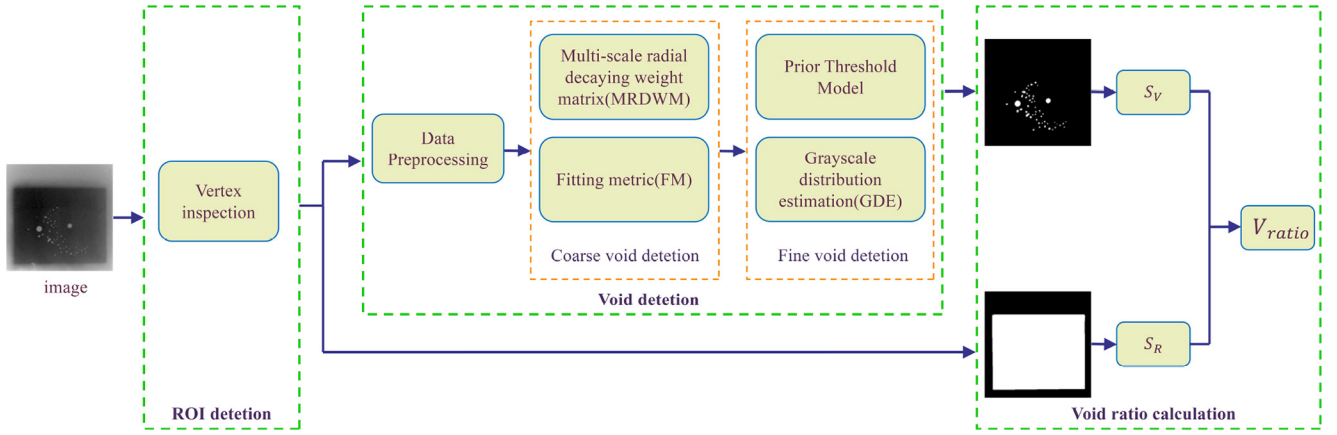


Fig. 3. The architecture of the proposed framework.

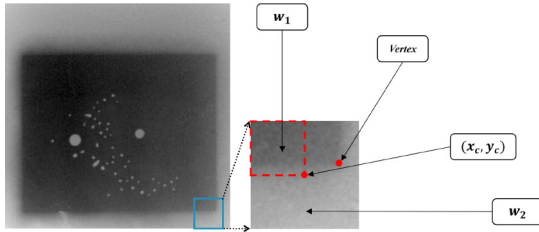


Fig. 4. Illustration of sliding window region decomposition.

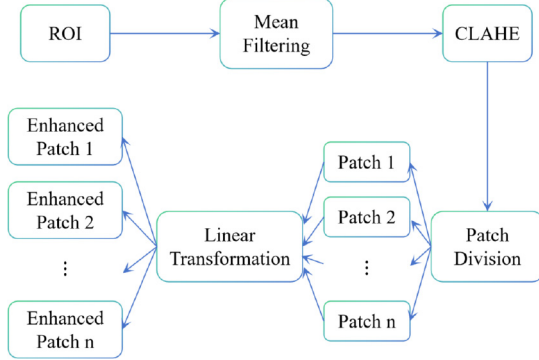


Fig. 5. The workflow of data preprocessing.

width to achieve the inspected patch. Thus, in this study, the neighbor patches have an overlap of 4-pixel width for each other. Finally, the inspected patch is processed by the simple linear transformation to further enhance the contrast. Fig. 5 shows the workflow of data preprocessing.

D. MRDWM-FM for Coarse Void Inspection

Since the voids in the MOSFET demonstrate various sizes and various irregular shapes, the MRDWM-FM combines a multi-scale matrix and a fitting metric to coarsely identify whether the patch involves the voids. Specifically, the multi-scale matrix W^s (s represents low scale l , medium scale m , or high scale h , correspondingly) is constructed as a radial decaying weight matrix to focus the inspection on the central regions of the voids, which can reduce the interferences of the void edges and noise and enhance the responses to the voids.

Its weight coefficient $\theta^s(x, y)$ at the point (x, y) is formulated as

$$\theta^s(x, y) = a^s(x, y) \times \varepsilon(a^s(x, y)) - \varepsilon(-a^s(x, y)), \quad s = l, m, h \quad (4)$$

$$a^s(x, y) = \frac{d^s}{2} - \sqrt{(x - x_0)^2 + (y - y_0)^2} \quad (5)$$

where d^s represents the size of the multi-scale matrix at the scale s , (x_0, y_0) denotes the center of the multi-scale matrix, and $\varepsilon(\cdot)$ represents the step function. The normalized weight coefficient $\omega^s(x, y)$ of W^s is formulated as

$$\omega^s(x, y) = \frac{\theta^s(x, y)}{\sum_{i=1}^{d^s} \sum_{j=1}^{d^s} \theta^s(i, j)} \quad (6)$$

Then, a set of windows $\{p_1^s, p_2^s, \dots, p_{n^s}^s\}$ can be obtained for a patch P by traversing this patch, where n^s represents the total number of windows. Next, to measure the void response within the patch, a response function $\Phi^s(P)$ is defined as

$$\Phi^s(P) = \frac{\max(\langle W^s, p_k^s \rangle)}{\mu}, \quad k = 1, 2, \dots, n^s \quad (7)$$

where $\langle \cdot \rangle$ represents the Frobenius inner product (FIP), and μ denotes the mean grayscale value of the patch P .

If the patch does not contain voids, the grayscale distribution within the patch approximates a Gaussian distribution, indicating that the difference between $\max(\langle W^s, p_k^s \rangle)$ and μ is relatively small to further make the response function $\Phi^s(P)$ relatively small. If the patch contains voids, the grayscale values of the void pixels are higher than those of the non-void pixels, indicating that the difference between $\max(\langle W^s, p_k^s \rangle)$ and μ is relatively larger to further make the response function $\Phi^s(P)$ relatively larger. The facts indicate that the response function $\Phi^s(P)$ can to some extent reveal the potential voids in the patch at multiple scales. To distinguish whether the patch contains voids, the response term $G_R(P)$ is defined as

$$G_R(P) = \varepsilon\left(\sum_s \Phi^s(P) - th_R\right) \quad (8)$$

where th_R represents the response threshold. $\sum_s \Phi^s(P)$ denotes the total response of the patch P at multiple scales. To determine the range of th_R , we conducted a statistical

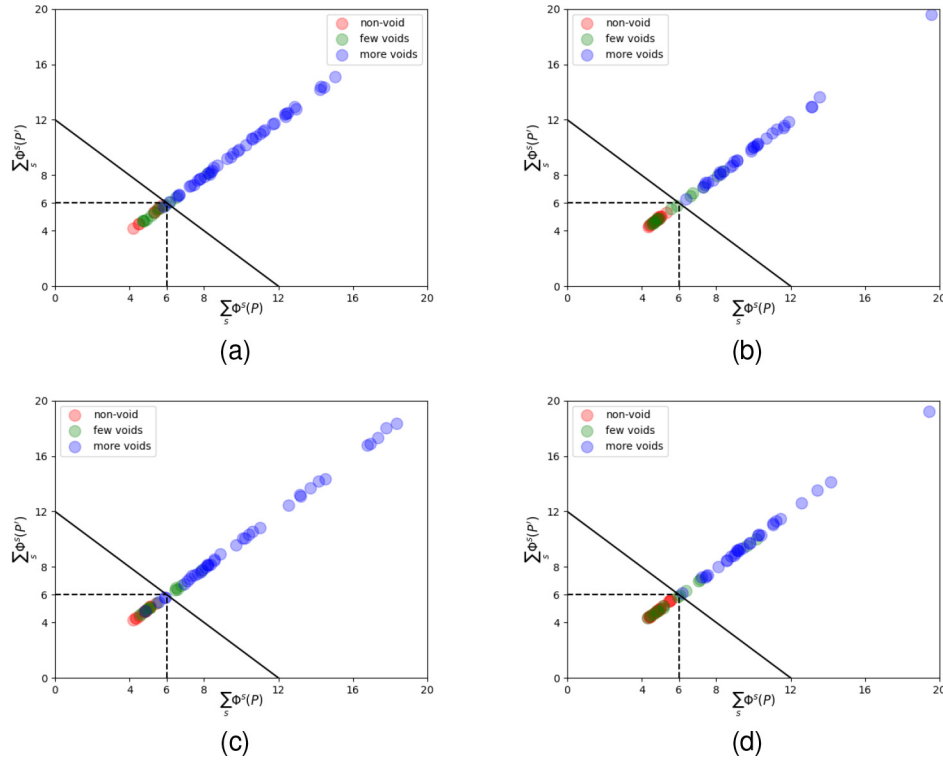


Fig. 6. Statistical analysis of the total responses of patches under different void levels.

analysis on the voids within a number of patches. To facilitate a more comprehensive statistical analysis, the patches were categorized into three levels: ‘non-void’, ‘few voids’ and ‘more voids’. Here, ‘non-void’ refers to the absence of void pixels within the patch P , ‘few voids’ indicates that the proportion of void pixels within P is between $(0, 0.01]$, and ‘more voids’ means that the proportion of void pixels within P exceeds 0.01. To enhance the stability of the statistical results and to reduce the interference of noise, we constructed a new patch as P' , in which each pixel value was calculated as the mean grayscale of the neighborhood of each pixel within P . Fig. 6 shows the intensity response of patches under different void levels, where the horizontal/vertical axes respectively represent $\sum_s \Phi^s(P)$ and $\sum_s \Phi^s(P')$, and the scatter points indicate various patches with/without voids. When $\sum_s \Phi^s(P) + \sum_s \Phi^s(P')$ is greater than 12, P is highly prone to be a void-containing patch. That is, whether the patch involves the voids is comprehensively determined by $\sum_s \Phi^s(P) + \sum_s \Phi^s(P')$. Thus, a simple but effective threshold optimization strategy is utilized to determine the range of th_R . Specifically, a line with the slope of -1 is translated across the scatter plot as indicated in Fig. 6. During translation, the two sums d_v and d_n of the distances of the five nearest patches with/without voids to the line are calculated. By ensuring that no more than five non-void patches lie above the line, minimizing d_v and d_n can adaptively derive the approximate range of th_R , which is around 6.

As illustrated in Fig. 6, the ‘non-void’ patch (red points) and the ‘few voids’ patch (green points) are not easily distinguishable. This is because some patches only contain a small number of small-sized voids, with relatively low $\sum_s \Phi^s(P)$.

However, the small-sized voids within these patches exhibit a stronger response at the low scale, while their responses at the other two scales are comparatively weaker. Thus, a scale difference term $G_C(P)$ is defined to reduce the risk of missing the identification of these patches, formulated as

$$G_C(P) = \varepsilon(D^m(P) + D^h(P) - th_C) \quad (9)$$

$$D^s(P) = \Phi^l(P) - \Phi^s(P), \quad s = m, h \quad (10)$$

where th_C represents the scale difference threshold. Fig. 7 shows the statistical analysis of different patches regarding the scale difference, where the horizontal/vertical axes represent $D^m(P)$ and $D^h(P)$, respectively. As indicated in (9), the identification is comprehensively influenced by $D^m(P) + D^h(P)$. Thus, the above threshold optimization strategy can be also conducted to determine the range of th_C . As indicated in Fig. 7, when th_C is empirically set to 0.8, ‘non-void’ patches and ‘few voids’ patches can be better distinguished.

Due to the influence of Gaussian noise in the X-ray images of MOSFETs, if a patch has no voids, its grayscale distribution approximates a Gaussian distribution. Thus, there is a large difference between the original grayscale distribution $O(P)$ of the patch P involving voids and the corresponding fitted one $Q(P)$ through maximum likelihood estimation. To evaluate this difference, the distribution term is defined as

$$G_D(P) = \varepsilon(D_{KL}(O(P)||Q(P)) - th_D) \quad (11)$$

$$D_{KL}(O(P)||Q(P)) = \sum O(P) \log\left(\frac{O(P)}{Q(P)}\right) \quad (12)$$

where th_D represents the distribution difference threshold. To obtain the approximate range of th_D , we conducted

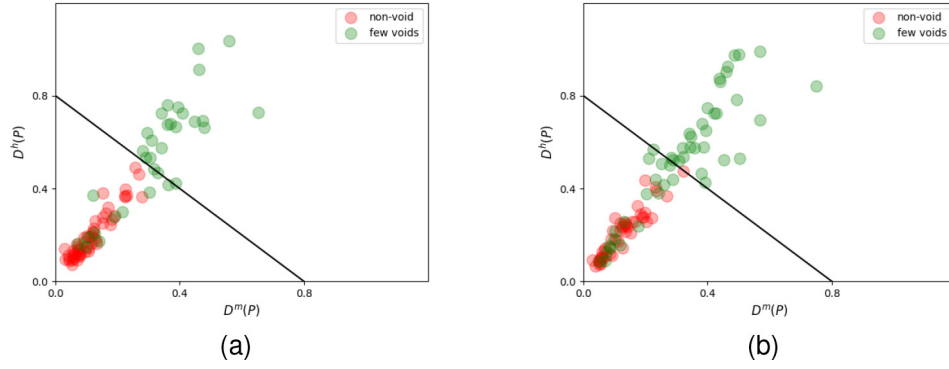


Fig. 7. Statistical analysis of the scale difference of patches under two void levels.

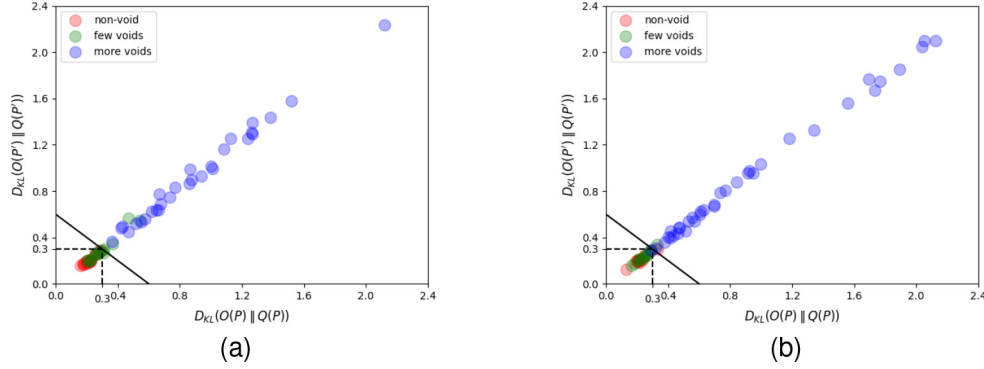


Fig. 8. Statistical analysis of the fitting metric of patches under three void levels.

a statistical analysis of the fitting metric. The horizontal/vertical axes of Fig. 8 represent $D_{KL}(O(P)||Q(P))$ and $D_{KL}(O(P')||Q(P'))$, respectively. Similarly, the identification of the patch classes is comprehensively influenced by $D_{KL}(O(P)||Q(P)) + D_{KL}(O(P')||Q(P'))$. Thus, the threshold optimization strategy is also utilized to determine the range of th_D . As illustrated in Fig. 8, when th_D is set to 0.3, the grayscale distribution can be more effectively utilized to distinguish void-containing patches from non-void patches.

Combining (8), (9) and (11), the identification function is defined as

$$G(P) = G_R(P) + G_C(P) + G_D(P) \quad (13)$$

If $G(P)$ is greater than 0, the patch P is identified as void-containing, otherwise as non-void.

E. GDE-AT for Fine Void Inspection

After the patches containing voids are identified by the MRDWM-FM, the proposed GDE-AT is utilized to finely inspect the voids in these patches, which is an adaptive thresholding scheme based on grayscale distribution estimation. Since the grayscale value distribution of the non-voids in the void-containing patch tends to approximate a Gaussian distribution, and the grayscale values of void pixels are typically higher than those of non-void pixels, the threshold th_{pi} for fine void inspection can be adaptively determined through the grayscale value distribution estimation of the non-voids, defined as

$$th_{pi} = \mu_e + \beta \times \sigma_e + \gamma \quad (14)$$

where the parameters μ_e and σ_e represent the mean and standard deviation of the grayscale values of the non-void pixels, β is the balancing factor, used to adjust the balance of μ_e and σ_e in th_{pi} , while γ is the correction term that adjusts th_{pi} when the grayscale distribution of the patch does not closely follow the expected Gaussian distribution.

Here, the grayscale distribution of the non-voids in the void-containing patch is utilized to estimate the above parameters. Assume f as the grayscale distribution of the void-containing patch, the cumulative frequency $F(z)$ of the interval centered at the grayscale value z is

$$F(z) = \sum_{t=-d}^d f(z+t) \quad (15)$$

where d determines the size of the interval. The center z_P of the interval with the highest cumulative frequency is formulated as

$$z_P = \arg \max F(z) \quad (16)$$

The grayscale distribution $f_n(t)$ of the non-voids can be formulated as

$$f_n(t) = f(t) \times \frac{N_p}{N_n}, t \in [p_{min}, 2 \times z_P - p_{min}] \quad (17)$$

where N_p represents the number of pixels in the patch, N_n denotes the number of pixels with grayscale values within $[p_{min}, 2 \times z_P - p_{min}]$, and p_{min} is the minimum grayscale value of the patch. Then, the mean μ_e and standard deviation σ_e of $f_n(t)$ can be obtained by maximum likelihood estimation.

β is responsible for adjusting the balance of μ_e and σ_e , which determine the definitions of void pixels and non-void pixels. Since the grayscale distribution of the non-voids approximates a Gaussian distribution, according to the 3-sigma rule, 99.7% of the data points are expected to fall within three standard deviations of the mean. Therefore, most of the pixel gray values in the non-voids will fall within this range, that is, the range of β is confined to [2, 4].

In real industries, since the grayscale distribution of non-voids in some patches does not fit well to a Gaussian distribution, z_P may deviate from the center of the non-void grayscale values, resulting that the interval $[p_{min}, 2 \times z_P - p_{min}]$ differs from the real non-void grayscale interval. This fact will result that th_{pi} is not the optimal threshold. To address this issue, a correction term γ is introduced to adjust the threshold th_{pi} , defined as

$$\gamma = \sigma_e \times r \times \varepsilon(r - \tau) \quad (18)$$

$$r = \frac{F(2 \times z_P - p_{min})}{F(z_P)} \quad (19)$$

where τ represents the activation threshold, empirically set to 0.5.

Afterward, the voids within the patch can be adaptively inspected by the threshold th_{pi} , formulated as

$$P_{lo}(m, n) = \begin{cases} 1, & P(m, n) \geq th_{pi} \\ 0, & P(m, n) < th_{pi} \end{cases} \quad (20)$$

where $P(m, n)$ is the grayscale value of the (m, n) -th pixel in the patch.

F. Pseudocode of the Proposed Framework

The pseudocode of the proposed framework is shown in Algorithm 1.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

All the experiments were performed on the personal computer with an 8-core Intel i7-11370H CPU @ 3.30GHz, 16GB of RAM, with PyCharm as the simulation platform.

Dice coefficient and mIoU are utilized to evaluate the inspection performance for the voids. It is noted that the higher the void ratio, the greater the potential performance risks of the device. Although different manufacturers have different QC standards, a 5% QC standard is a relatively strict criterion that represents the empirical level in the power device manufacturing [3], which is adopted by many manufacturers. Thus, this strict QC standard of 5% is utilized for determining the qualification of a MOSFET. Accuracy and F1-score are utilized to evaluate the inspection performance for the MOSFETs, and the chi-square value χ^2 and p-value are introduced to assess the statistical significance of inspection results in comparison with random guessing. Additionally, the processing time per sample (s/sample, i.e., processing speed) is used to evaluate the time efficiencies of different methods.

A. Influence of Preprocessing Parameters

To validate the rationality of the preprocessing parameters, we evaluated the inspection performance under different numbers of patches and numbers of overlapping pixels. Meanwhile,

Algorithm 1 Proposed Framework

Input: X-ray image of MOSFET X_I

Output: The void ratio of MOSFET V_{ratio}

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1:  $R \leftarrow$  ROI detection( $X_I$ )
2:  $R_m \leftarrow$  Mean filtering( $R$ )
3:  $R_c \leftarrow$  CLAHE( $R_m$ )
4:  $\{P_1, \dots, P_n\} \leftarrow$  Patch Division( $R_c$ )
5:  $\{P_1^f, \dots, P_n^f\} \leftarrow$  Linear Transformation( $\{P_1, \dots, P_n\}$ )
6: Patches with voids  $P_{list}^v \leftarrow \{\}$ 
7: for  $P_i^f$  in  $\{P_1^f, \dots, P_n^f\}$ :
  // As indicated in (13)
  if  $G_R(P_i^f) + G_C(P_i^f) + G_D(P_i^f) = 0$ :
    continue
  else:
     $P_{list}^v \leftarrow$  expand( $P_i^f$ )
  end
8: The void segmentation result  $P_{list}^v \leftarrow \{\}$  in the patch
9: for  $P_i^v$  in  $P_{list}^v$ :
   $P_{list}^v \leftarrow$  expand(GDE-AT( $P_i^v$ ))
  end
10:  $X_I$ 's void inspection result  $X_v \leftarrow$  Stitching( $P_{list}^v$ )
11:  $S_R \leftarrow$  Pixel Calculation( $X_R$ )
12:  $S_V \leftarrow$  Pixel Calculation( $X_v$ )
13:  $V_{ratio} \leftarrow S_R \text{ and } S_V$  // As indicated in (1)

```

TABLE I
THE INFLUENCE OF THE NUMBER OF PATCHES ON INSPECTION RATE FOR SMALL VOID PATCHES AND INSPECTION PERFORMANCE FOR THE ROI.

Number of patches	1×1	2×2	3×3	4×4	5×5
Dice	0.6990	0.7924	0.8852	0.8626	0.8536
Inspection rate(%)	/	89.94	90.42	88.77	87.92

to evaluate the impact of different parameters on small voids, we define a “small void patch” as a patch where the proportion of void pixels (relative to the total number of pixels in the patch) falls within the range of (0, 0.01].

TABLE I presents the inspection rate for small void patches and the inspection performance for the ROI (i.e., Dice) under different numbers of patches. When the number of patches is 1×1, there is no need for void coarse inspection, and thus the inspection rate of small void patches is not calculated. With the increase of the number of patches, the Dice and the inspection rate first increase and then decrease, reaching their maxima when the number of patches is 3×3. Therefore, the number of patches in this study is empirically set to 3×3.

TABLE II presents the inspection performance for the ROI and inspection rate for small void patches under different numbers of boundary overlapping pixels. As the number of overlapping pixels increases, the inspection rate for small void patches gradually arises, which indicates that a large number of overlapping pixels enhances the sensitivity to small voids. However, a large number of overlapping pixels implies an increase of the patch size, which further amplifies the adverse impact caused by the uneven gray-scale distribution of the ROI. Moreover, the contribution of the high inspection rate

TABLE II
THE INFLUENCE OF THE NUMBER OF OVERLAPPING PIXELS ON
INSPECTION PERFORMANCE FOR THE ROI AND INSPECTION RATE FOR
SMALL VOID PATCHES.

Number of overlapping pixels	2	4	6	8	10
Dice	0.8847	0.8852	0.8838	0.8825	0.8817
Inspection rate(%)	88.83	90.42	92.78	93.68	94.50

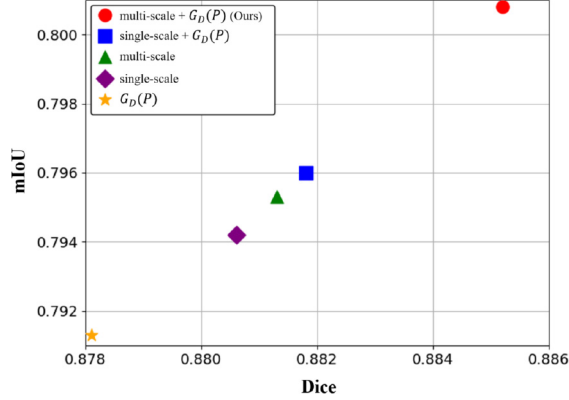


Fig. 9. Inspection results achieved by the proposed models with different coarse void inspection schemes.

for small void patches to the overall inspection performance is very limited. This leads to a gradual decline in the Dice score when the number of overlapping pixels exceeds 4. Therefore, the number of overlapping pixels in this study is empirically set to 4.

B. Discussion on Coarse Void Inspection

To evaluate the influences of the multi-scale scheme and the fitting metric in the MRDWM-FM, we conducted an experiment to compare the void inspection performance achieved by the proposed models with different coarse void inspection schemes. Specifically, “ $G_D(P)$ ” represents that the inspection model only utilizes the distribution term for coarse void inspection. “single-scale” and “multi-scale” denote that the inspection models only utilize the response term at a single scale for coarse void inspection, and the combination of the response term and the scale difference term at three scales, respectively.

As indicated in Fig. 9, the “ $G_D(P)$ ” model performs the worst void inspection performance, since the fitted grayscale distribution of the patch containing many voids with small sizes is quite similar to the original one. The “multi-scale” model performs better than the “single-scale” model, since the multi-scale scheme can better capture the size and shape characteristics of the voids with various sizes and various irregular shapes. When the multi-scale scheme is combined with distribution term $G_D(P)$, the corresponding model, i.e., “multi-scale + $G_D(P)$ ” performs the best void inspection performance among all the models.

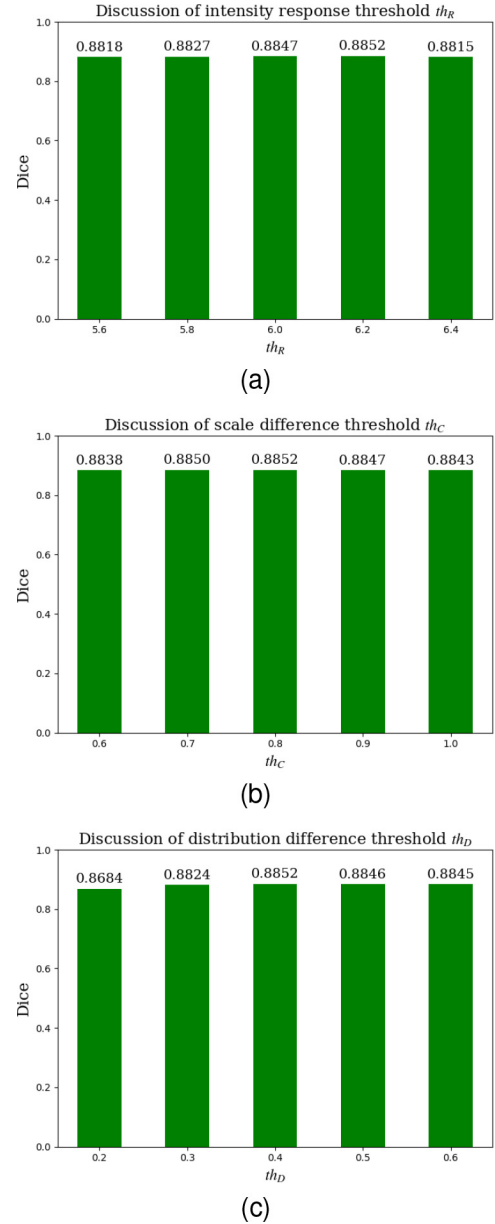


Fig. 10. Inspection results achieved by our inspection method with different thresholds in the MRDWM-FM.

C. Discussion on the Thresholds of MRDWM-FM

As indicated in (8), (9) and (11), three key parameters in MRDWM-FM significantly determine the three terms of the identification function in (13), involving intensity response threshold th_R , scale difference threshold th_C , and distribution difference threshold th_D . That is to say, the three thresholds should be elaborately set to achieve a good inspection performance for MOSFETs. To decrease the time-consuming attempts to find these optimal thresholds, we provided a threshold optimization strategy to constrain the attempts into some value ranges, as mentioned in Section III-D. Here, an experiment was conducted to validate this strategy. As indicated in Fig. 10, if each threshold falls into the range determined by the threshold optimization strategy, our method can achieve comparable void inspection performance. That is,

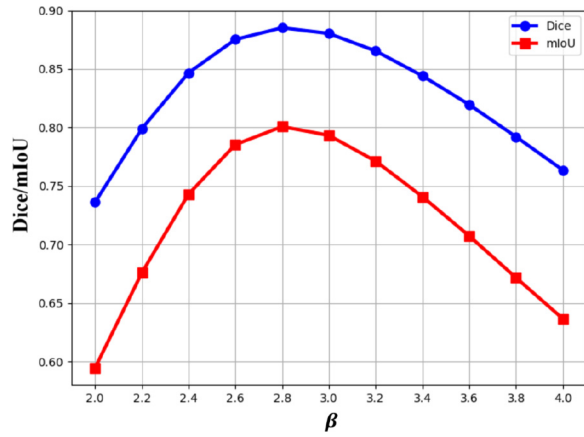


Fig. 11. Inspection performance achieved by the inspection models with different β .

our method is relatively insensitive to the thresholds within these ranges. In our study, th_R , th_C and th_D are empirically set to 6.2, 0.8, 0.4, correspondingly. In industrial applications, a small portion of samples can be employed to conduct this strategy, which is highly simple and operable.

D. Influence of β

In the GDE-AT, the balancing factor β determines the threshold th_{pi} for fine void inspection. Fig. 11 shows the void inspection performance achieved by the inspection models with β . As β increases, both of Dice coefficient and mIoU first arise and then gradually decrease. This is because the smaller β is, the smaller the threshold th_{pi} is, resulting that the more non-void pixels are misclassified as void pixels. In contrast, the larger β is, the larger the threshold th_{pi} is, resulting that the more void pixels are misclassified as non-void pixels. When β is 2.8, both metrics reach their maxima, which is consistent with the discussion on the range of β in Section III-E.

E. Influence of γ

To demonstrate the influence of the correction factor γ , we conducted an experiment to evaluate the void inspection performance achieved by the proposed method with/without γ . As indicated in TABLE III, the proposed method with γ performs a little better than that without γ , indicating that γ can to some extent deal with the problem of non-Gaussian grayscale distribution for some non-voids.

To visually illustrate the influence of γ , three patch samples (labeled by red boxes) were selected for void inspection, in which the grayscale values of non-voids do not conform the Gaussian distribution. As indicated in Fig. 12, there is quite difference between the original distribution of the grayscale values and the fitted Gaussian distribution for the non-voids in each patch. If the proposed method performs void inspection without γ , some non-voids are wrongly inspected as voids. The visual results are consistent with the statistical analysis in TABLE III.

TABLE III
INFLUENCE OF γ ON VOID INSPECTION.

	With γ	Without γ
Dice	0.8852	0.8801

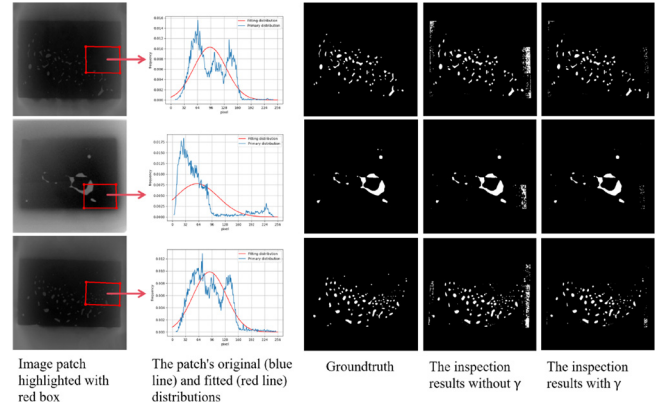


Fig. 12. Visualization of void inspection with/without γ .

TABLE IV
ABLATION STUDY ON KEY STAGES OF THE PROPOSED METHOD.

Data Preprocessing	MRDWM-FM	GDE-AT	Dice
			0.1505
✓			0.1823
		✓	0.3120
✓	✓		0.3148
✓		✓	0.8546
✓	✓	✓	0.8852

F. Ablation Study

To validate the stages of data preprocessing, MRDWM-FM and GDE-AT, we conducted an ablation experiment to perform void inspection via the proposed method with different stages. It is noted that, since MRDWM-FM functions on the patch level, it is not implemented if data preprocessing stage is absent. When GDE-AT is absent, OTSU is used as a substitute to finely inspect voids. That is, the baseline refers to the OTSU method for void inspection.

As shown in TABLE IV, if the GDE-AT is absent, the baseline method and the baseline methods with data processing and/or MRDWM-FM achieve almost the three worst void performance, demonstrating the significance of the GDE-AT for pixel-level void inspection. When only the GDE-AT is used for fine void inspection, the uneven grayscale distribution in the ROI causes some non-void pixels to have higher grayscale values than void pixels, resulting in poor inspection performance with 0.3120 Dice. When the baseline is equipped with data preprocessing and GDE-AT, the inspection method achieves the second best void inspection performance with 0.8546 Dice. This is because the grayscale distribution of non-void patches tends to follow a Gaussian distribution to result in a large threshold th_{pi} for the GDE-AT, which reduces false positives to some extent. However, the non-voids are potentially mis-inspected by the GDE-AT as voids if the

TABLE V
STATISTICAL VALIDATION FOR OUR METHOD.

Fold No.	Number of samples	Number of qualified samples	Number of unqualified samples	χ^2	p-value	Accuracy(%)	F1-score
Fold 1	104	82	22	13.57	<0.001	96.15	0.9759
Fold 2	104	91	13	5.29	<0.025	98.08	0.9890
Fold 3	104	86	18	10.39	<0.001	98.08	0.9882
Fold 4	104	78	26	18.85	<0.001	96.15	0.9737
Fold 5	103	79	24	17.28	<0.001	97.09	0.9811

TABLE VI
STATISTICAL RESULTS OF VOID INSPECTION VIA DIFFERENT METHODS.

Methods	Categories	Dice	Accuracy(%)	F1-score	χ^2	p-value	Speed(s/sample)
SP-IGA-ACM(2023) [25]	Image Processing-based	0.3093	80.15	0.8898	10.94	9.41×10^{-4}	25.499
MC-ACK(2023) [25]	Image Processing-based	0.3722	80.73	0.8853	12.02	5.26×10^{-4}	1.615
LPF(2019) [27]	Image Processing-based	0.5589	76.88	0.8522	6.03	1.41×10^{-2}	27.976
LoG(2012) [17]	Image Processing-based	0.5479	76.69	0.8614	5.52	1.88×10^{-2}	0.070
Xie et al.(2019) [28]	Image Processing-based	0.5433	76.30	0.8200	5.03	2.49×10^{-2}	0.874
HSH-UNet(2024) [29]	Deep Learning-based	0.8808	96.14	0.9763	60.41	7.70×10^{-15}	3.013
ID-UNet(2025) [30]	Deep Learning-based	0.8687	95.32	0.9709	56.35	6.06×10^{-14}	2.102
Ours	Image Processing-based	0.8852	97.11	0.9819	63.76	1.41×10^{-15}	0.812

MRDWM-FM is absent to distinguish these non-voids. So, the proposed method with the three stages achieves the best void inspection performance of 0.8852 Dice.

G. Statistical Validation

To evaluate the robustness of our method, we conducted statistical validation on the experimental results of the proposed method. To do so, our dataset was evenly divided into 5 folds through random allocation. Due to the randomness, the number of qualified and unqualified samples in each fold was inconsistent. Subsequently, the chi-square test was employed to derive the chi-square statistic χ^2 and corresponding p-value for each fold. Accuracy and F1-score for each fold were then computed to evaluate the model's performance.

As shown in TABLE V, the p-value for Fold 2 is <0.025, while those for the other four folds are all <0.001. This is due to the randomness of the dataset splits: more qualified samples in Fold 2 result in a smaller chi-square statistic χ^2 , making its p-value slightly larger than those of the other folds. The proposed method for all the folds shows statistically significance with the largest p-value <0.025, confirming that it is robust against variations in data splits and class imbalance.

Compared with Accuracy, F1-score demonstrates slightly more robust across the 5 folds, with a standard deviation of only 0.0062, which is slightly smaller than Accuracy's standard deviation of 0.86%. And the performance fluctuations across the 5 folds (Accuracy: 96.15%-98.08%, F1-score: 0.9737-0.9890) are attributed to random errors, indicating that our method is insensitive to data splits and exhibits a certain stability.

H. Comparison With Other Methods

Since our proposed method is image processing based inspection method rather than deep learning based one, we mainly compared it to several existing image processing based methods, such as LoG [17], LPF [27], Xie et al. [28], SP-IGA-ACM [25], and MC-ACK [25]. Due to the popularity of deep learning, we also compared it to two deep learning based methods, such as HSH-UNet [29] and ID-UNet [30]. TABLE VI presents the statistical analysis of void inspection via different methods.

SP-IGA-ACM [25] is an active contour method based on shape priors and average grayscale strategy, in which the contours are guided to the void boundaries for void segmentation by the dynamic evolution of contours. However, due to its reliance on circular shape priors, a large number of non-circular voids in the MOSFETs are not effectively inspected. Additionally, for the voids with weak edges, it struggles to guide the contour to converge to their real boundaries. Since a large number of voids are missed by inspection, it achieves the lowest Dice coefficient of 0.3093 among all the methods, which is unacceptable in real industries.

The MC-ACK [25] preprocesses the image using the proposed multi-scale circular convolution kernels and then inspects voids by combining OTSU and adaptive thresholding. However, the fixed circular shape of the convolution kernels limits the response to non-circular voids, resulting in ineffective enhancement of non-circular voids. The fact makes its subsequent segmentation cannot well inspect non-circular voids, resulting in a poor inspection performance of 0.3722 Dice.

The LoG method [17] preprocesses images through Gaussian blurring and then detects void edges via the

Laplacian operator (i.e., a second-order derivative). Since the second-order derivative is not sensitive to the subtle grayscale differences between the voids and non-voids, the LoG method struggles to detect the voids with weak edges, resulting in a low Accuracy of 76.69%. Due to its simple processing mechanism, its inspection speed is the fastest among all the methods, reaching 0.070 s/sample, which is welcome for real industrial inspection.

The LPF method [27] makes the contour converge to the void boundary for void inspection, which is implemented by complex energy function calculations and iterative optimization process. These complex energy function calculations and iterative optimization not only make the LPF method difficult to inspect the voids with weak edges, but also result in prolonged inspection time. Specifically, it costs 27.976s to inspect a MOSFET in average, which is the slowest speed among all the methods and unacceptable for real industrial applications.

The method proposed by Xie et al. [28] calculates extreme values using the grayscale frequency histogram to narrow the threshold range, screens thresholds based on the principle of maximum inter-class variance, and finally inspects voids using the selected thresholds. However, extreme values and grayscale variance are merely simple data features, which fail to fully utilize the grayscale distribution information of MOSFET X-ray images and is easily affected by noise. Thus, it achieves the worst Accuracy of 76.30% among all the methods.

Since the deep learning method HSH-UNet [29] embeds a hybrid selective high-order interaction module HSHB into the traditional U-Net network for segmentation, it performs excellently in inspection for both voids and MOSFETs, and only slightly inferior to our proposed method. However, its complex hierarchical design with excessive network layers leads to redundant computations to decrease inspection efficiency, with the inspection speed of 3.013 s/sample. Compared with HSH-UNet, ID-UNet [30] shows a slight improvement in inspection efficiency, with an inspection speed of 2.102 s/sample. However, its complex cross structures and redundant computations still limit its time efficiency, whose inspection speed is slower than our proposed method. Moreover, since ID-UNet embeds a feature extraction module named Infrared Small Target Feature Extraction (ISTFE) into the U-Net architecture, it places more emphasis on small-scale targets, resulting that it only achieves the third best inspection performance for voids and MOSFETs with 0.8687 Dice, 95.32% Accuracy and 0.9709 F1-score. Although the two deep learning based methods, i.e., HSH-UNet and ID-UNet, achieve fairly good inspection performance with relatively fast inspection speed, the inherent drawbacks of deep learning, such as time-consuming manual data annotation and model training, limit their real industrial applications to some extent. Moreover, deep learning models consume substantial memories and storage capacities [31], and their irregular memory access patterns prevent GPUs from fully leveraging their hardware advantages [32]. Due to the inherent dynamic nature of deep learning models, even when accelerated with static programming languages (e.g., C++), deep learning still faces with the challenging to achieve significant improvements in

inference speed. These factors further hinder their deployment in real-world industrial applications.

Comparatively, our proposed method achieves the best inspection performance among all the methods at the second fastest speed of 0.812 s/sample, with the performance of 0.8852 Dice, 97.11% Accuracy, and 0.9819 F1-score. Moreover, the p-value for our method is only 1.41×10^{-15} , which is slightly superior to those for the two deep learning based methods yet significantly better than those for image processing based methods. This analysis indicates that our method exhibits an extremely high degree of statistical significance compared to random guessing. This can be attributed to two novel schemes, i.e., MRDWM-FM and GDE-AT. In the former scheme, the constructed multi-scale radial decaying weight matrix can extract the intensity and scale features from the patches at different scales, while the fitting metric can extract the grayscale distribution features from the patches. The three types of features can well characterize the patches with/without voids. The latter scheme makes full advantages of the grayscale values of the non-voids to construct an adaptive threshold function, which can well distinguish the voids and the non-voids in the void-containing patches. Moreover, in contrast to deep learning methods, our method can be easily accelerated by some parallel strategies since it is an image processing based method, which can exhibit more structured dataflow and stronger static properties. This indicates that our method can be further accelerated in the future to satisfy the real-time inspection requirement in real industrial scenarios.

I. Visualization of Inspection Results

Fig. 13 shows the visual inspection results for two qualified MOSFETs and two unqualified MOSFETs, with the QC standard of 5%. As shown in Fig. 13, SP-IGA-ACM and MC-ACK can hardly inspect non-circular voids, resulting in very low void ratios calculated by them compared to the ground-truth ones. The facts further result that many unqualified MOSFETs are misclassified as qualified. The LoG method fails to find many voids with highly-irregular contours. The LPF method and the Xie et al. method perform some over-inspections in some MOSFETs. Our inspection method, along with the deep learning methods (HSH-UNet and ID-UNet), precisely inspect all the voids in the MOSFETs. Notably, the void ratios calculated by our method are slightly closer to the ground-truths compared to HSH-UNet and ID-UNet. The visual results are consistent with the above objective statistical analysis in Section IV-H.

J. Error Analysis and Visualization

Fig. 14 shows two mis-inspection cases for our proposed method, including one false positive (FP, an unqualified sample mis-inspected as qualified) and one false negative (FN, a qualified sample mis-inspected as unqualified). As shown in Fig. 14(a), some non-voids in the FN sample are misjudged as voids, leading to an excessively void ratio in this sample. Thus, this sample is mis-inspected as unqualified. This is because when the grayscale distribution of non-voids within the patch does not closely follow a Gaussian distribution, GDE-AT

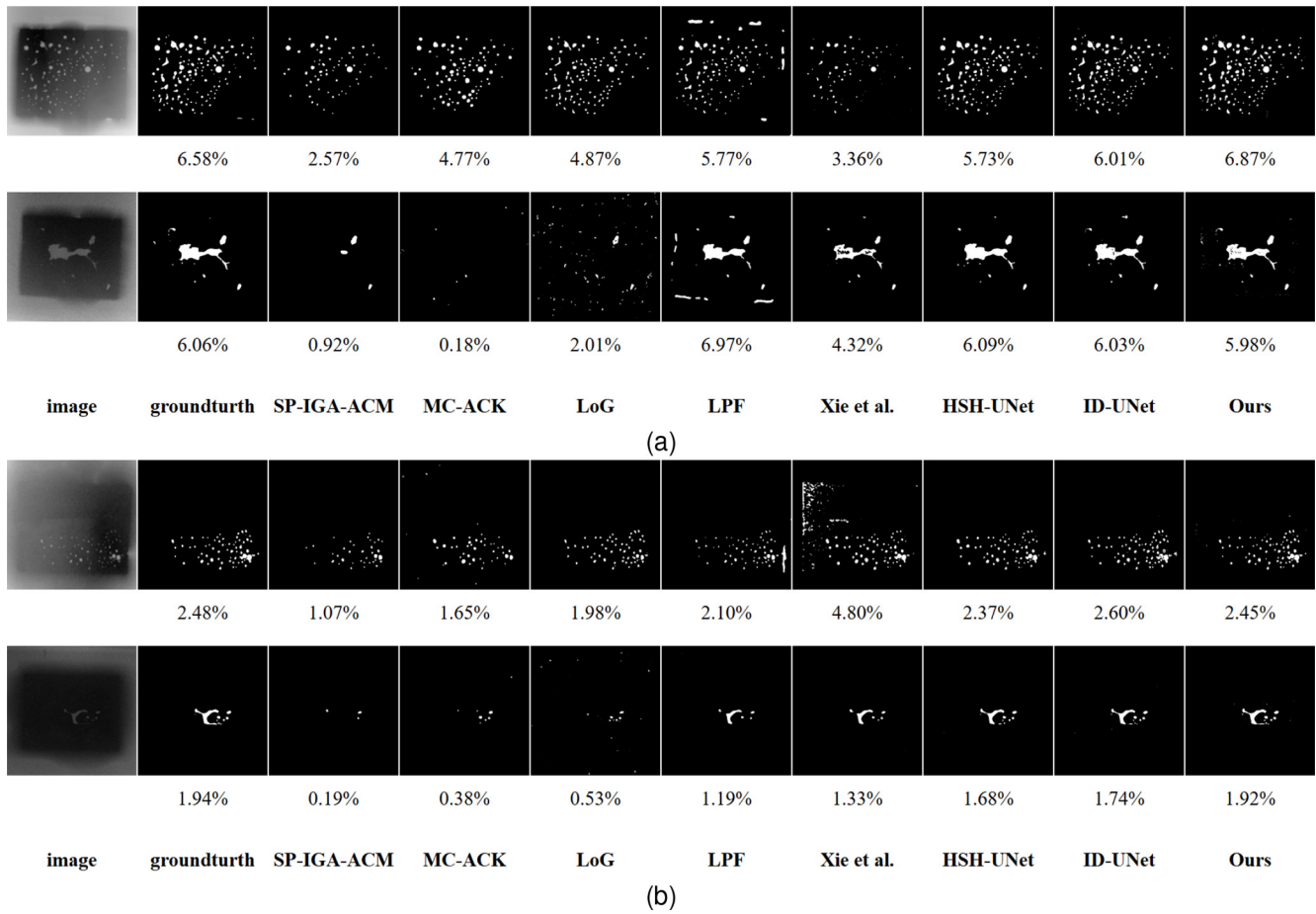


Fig. 13. Visualizations of void inspection results via different inspection methods. (a) Unqualified samples. (b) Qualified samples.

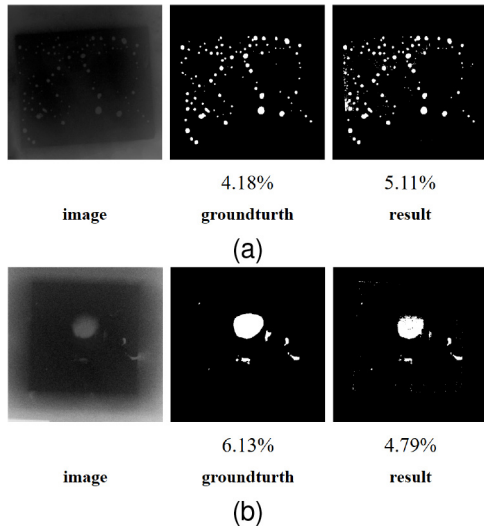


Fig. 14. Visualization results of mis-inspection cases. (a) FN. (b) FP.

exhibits insufficient adaptability to the grayscale distribution. For the FP sample in Fig. 14(b), the edges of large-scale voids are affected by noise, causing many void pixels to be misjudged as non-void pixels. This leads to the mis-inspection of the unqualified sample as qualified.

According to the analysis of the two cases, our method exhibits suboptimal performance when facing complex grayscale distributions and serious noise interference, even though it involves a correction factor in GDE-AT. This limitation can be possibly addressed by introducing a dual-threshold adaptive thresholding method to adapt to complex grayscale distributions in the future.

Severe lighting variation is another issue to influence the inspections. Fig. 15 presents two inspection cases under bad lighting conditions. As indicated in Fig. 15, the grayscales of void edges are very close to those of non-voids, causing some void pixels to be misclassified by fine inspection as non-void ones. In the future, we will collect more samples under various lighting conditions to develop a more robust dual-threshold adaptive method for fine inspection, which can alleviate this unwelcome issue to a certain extent owing to its inherent ability to perceive edges.

K. Validation on Additional Samples

To validate the generalization ability of the proposed framework, we collected another batch of MOSFETs involving 61 samples from the manufacturer to construct a new validation dataset. We directly employed our inspection framework without any parameter adjustments on these MOSFET samples. As indicated in TABLE VII, the proposed framework

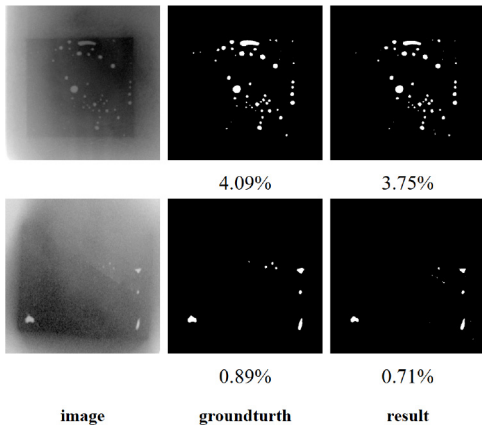


Fig. 15. Visualization of inspections under severe lighting conditions.

TABLE VII
STATISTICAL RESULTS ON THE ADDITIONAL SAMPLES

Dice	Accuracy(%)	F1-score	Speed(s/sample)
0.8670	98.36	0.9895	1.004

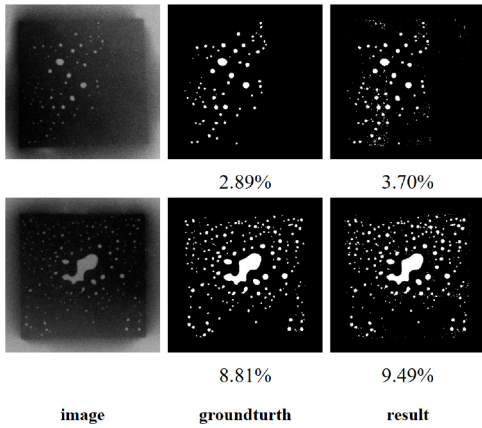


Fig. 16. Visualization results of two samples in the new dataset.

also performs a fairly good inspection performance for this new dataset.

Although the proposed framework shows an acceptable ability to judge sample eligibility in the new dataset, its Dice has dropped from 0.8852 for the original dataset to 0.8670 for the new dataset. Fig. 16 presents two examples of inspection results in the new dataset. As indicated in Fig. 16, some over-inspections emerge in both samples. This can be attributed to the stronger noise interference in the new dataset, coupled with a greater inhomogeneity in the grayscale distribution of the ROI, which increase the complexity of grayscale distribution within patches to damage the assumption of Gaussian distribution in our inspection framework.

L. Limitations and Future Work

Although our proposed method has achieved a promising performance for void inspection in MOSFETs at a reasonable speed, some issues are still considered as follows.

The ranges of the three thresholds in the MRDWM-FM can be determined by a simple but effective threshold optimization strategy. Although our method is relatively insensitive to the thresholds within the ranges, the optimal thresholds should be empirically set to achieve excellent inspection performance in our study. When faced with different packaging types or process conditions, the three thresholds can be empirically determined by this optimization strategy. However, to improve the generalization and the inspection performance, it is better to adaptively determine these optimal thresholds for different packaging types or process conditions. To this end, we can offline and adaptively determine the optimal thresholds by the grid-searching strategy within the ranges of the thresholds in the future.

Although the correction factor can to some extent alleviate the influence of the assumption of the Gaussian distribution for non-voids on void inspection, only 0.51% improvement is achieved, indicating that this assumption greatly influences the optimal determination of the threshold th_{pi} for fine void inspection. In the future, we will design an adaptive dual-threshold method based on local optimization to enhance the adaptability to diverse grayscale distributions, which first determines the approximate threshold range and then refines the optimal thresholds by weighting data features around candidate thresholds within the threshold range.

Although our method achieves an acceptable inspection speed of 0.812 s/sample, which is implemented by Python and on CPU, some improvements of its inspection speed can be conducted in the future to well adapt it to real industrial deployment. For example, we can use C++ programming to optimize key computational modules and reduce redundant memory operations. We can also refactor compute-intensive tasks based on the CUDA or OpenCL framework to leverage the multi-core architecture of GPUs.

Moreover, internal void inspection is not only significant for product QC, but also can be used to follow up the causes of void formation. If our method is deployed in real industries, a large number of MOSFETs with voids will have been inspected for a long period. We can analyze these anomaly MOSFETs to correlate manufacturing process parameters with void characteristics to follow up on the malfunction causes of void generation in the manufacturing process, which is beneficial to timely address potential manufacturing faults and to even improve the manufacturing process.

V. CONCLUSION

The internal voids in MOSFETs affect the electrical performance of the MOSFETs to further influence the reliability and lifespan of the electronic device. In this article, a multi-scale adaptive inspection framework is proposed to automatically inspect the voids in MOSFETs through the X-ray imaging, mainly consisting of ROI detection, void detection, and void ratio calculation. In the ROI detection stage, a location-aware vertex detection scheme is proposed for the transistor region detection. In the void detection stage, coarse-to-fine void inspection is performed on the detected

ROIs of the MOSFETs, involving the designed MRDWM-FM for coarse void inspection and the designed GDE-AT for fine void inspection. Finally, the void ratio for a MOSFET is calculated according to its detected ROI and inspected voids.

Experimental results demonstrate that the constructed multi-scale matrix and the fitting metric contribute to coarse void inspection, as well as that the balancing factor in the GDE-AT scheme influences the adaptive segmentation threshold for fine void inspection. Comparison experiments indicate that our inspection method is superior to some existing methods at an acceptable inspection speed of 0.812 s/sample, in terms of 0.8852 Dice, 97.11% Accuracy, and 0.9819 F1-Score.

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